MODULE 11 · DOMAIN 2 - IMPLEMENT GENERATIVE AI AND AGENTIC SOLUTIONS

Connecting an app to Foundry

The project client, the v1 API, the Responses API, conversation state, multimodal input and connections.

DOMAIN 2 - Generative AI and agents WEIGHT 30-35% MODULE 11 of 39

admin@sthithapragna.com

A call works when three things line up: the **right endpoint**, a **credential with the right role**, and the **deployment name**.

Connecting an app to a Foundry project

01
Install
azure-ai-projects 2.x and azure-identity.

02
Endpoint
The project endpoint URL.

03
Credential
DefaultAzureCredential, not a key.

04
Project client
AIProjectClient.

05
OpenAI client
From the project client.

06
Call
Responses API, model set to a deployment name.

ON THE EXAM

The same code runs on a laptop and in Azure: DefaultAzureCredential picks up a developer sign-in or a managed identity.

The minimal Foundry call

```
from azure.ai.projects import AIProjectClient
from azure.identity import DefaultAzureCredential

project = AIProjectClient(
    endpoint="https://contoso.services.ai.azure.com/api/projects/support",
    credential=DefaultAzureCredential())
client = project.get_openai_client()

response = client.responses.create(
    model="chat-prod",   # deployment name
    input="Summarise this ticket.")
print(response.output_text)
```

ON THE EXAM

Three highlighted lines, three failure points: the endpoint, the credential, and the deployment name.

Clients and endpoints, revisited

USE	FOR
`Project client (AIProjectClient)`	Connections, project settings, tracing setup
`OpenAI client from the project`	Responses, agents, evaluations, platform tools
`OpenAI SDK on /openai/v1`	Embeddings, maximum OpenAI compatibility
`Anthropic SDK on /anthropic`	Claude models deployed in Foundry
`Foundry Tools SDKs`	Vision, Speech, Language, Translator, Content Safety

ON THE EXAM

Module 01's exception still applies: the project endpoint doesn't route embedding requests. Use /openai/v1.

No more dated API versions

- The **v1 API** removes the dated **api-version** parameter that older code passed on every call.
- Base URL: **https://RESOURCE.openai.azure.com/openai/v1/** or the services.ai.azure.com form.
- The standard OpenAI client supports **token authentication with automatic refresh**.
- Preview features are switched on with **headers**, not by changing API versions.
- Chat Completions calls also work with some other providers' models, such as DeepSeek and Grok.

> ON THE EXAM
>
> Code that still sets api_version='2024-02-15-preview' is using the old pattern. New features need the v1 API.

Keyless OpenAI SDK on the v1 endpoint

```
from openai import OpenAI
from azure.identity import DefaultAzureCredential, get_bearer_token_provider

token_provider = get_bearer_token_provider(
    DefaultAzureCredential(), "https://ai.azure.com/.default")
client = OpenAI(
    base_url="https://contoso.openai.azure.com/openai/v1/",
    api_key=token_provider)

vectors = client.embeddings.create(
    model="embed-small", input=["refund policy"])
```

ON THE EXAM

api_key receives a token provider, not a secret. A 401 here often means the wrong token scope.

Responses or Chat Completions?

RESPONSES API

Recommended for new work

- Can keep conversation state for you
- Built-in tools: MCP, code interpreter, image generation
- Structured output set in text.format
- Guardrail results in a content_filters array

CHAT COMPLETIONS

Still supported

- You resend the whole message history
- Tools you implement yourself
- Structured output set in response_format
- Filter results in content_filter_results

THE TIEBREAKER

Microsoft recommends the Responses API for Azure OpenAI models. MCP tools work only in the Responses API.

Four ways to carry context between turns

APPROACH	HOW IT WORKS	CHOOSE IT WHEN
`previous_response_id`	The service stores turns; you pass the last ID	Simplest multi-turn chat
`Manual input array`	You append output items to the next input	You want full control of context
`store=false, encrypted reasoning`	Nothing stored; reasoning passed back encrypted	The service must not retain data
`Compaction`	Shrinks long context while keeping state	Very long conversations

ON THE EXAM

Stored responses are retained for 30 days by default, and can be deleted by ID.

'The service must not store conversations' means **store=false**. And that rules out **background mode**, which needs storage.

Streaming and background tasks

- **stream=True** sends text as it's generated, as output_text.delta events.
- A failure mid-stream arrives as an **error event inside an HTTP 200** response.
- Error events map to familiar codes: too_many_requests is 429, server_error is 500.
- You are **not charged** for tokens in a streamed response that fails.
- **background=True** queues long reasoning tasks: poll for status, or cancel.

> WATCH FOR THIS
>
> Monitoring only HTTP status codes will miss streaming failures. Handle error events in the stream itself.

Function calling with the Responses API

```
tools = [{"type": "function", "name": "get_order",
          "parameters": {"type": "object", "properties": {"order_id": {"type": "string"}}}}]
first = client.responses.create(model="chat-prod",
    tools=tools, input="Where is order 881?")
outputs = []
for item in first.output:
    if item.type == "function_call":
        result = lookup_order(json.loads(item.arguments))
        outputs.append({"type": "function_call_output",
            "call_id": item.call_id, "output": json.dumps(result)})
final = client.responses.create(model="chat-prod",
    previous_response_id=first.id, input=outputs)
```

ON THE EXAM

The model only requests the call. Your code runs it and returns the result, matched by call_id.

Sending images and PDFs

- **Images** as input_image: a public URL, a base64 data URI, or an uploaded file ID.
- **detail** can be low, high or auto. Low uses fewer tokens but can miss fine detail.
- Up to 50 images per request, with the images totalling under 50 MB.
- **PDFs** as input_file put both the extracted text **and an image of every page** into context.
- Only models that accept both text and images can take PDF input.

> ON THE EXAM
>
> A long PDF sent to a vision model is expensive. For fixed fields, a purpose-built extractor is often the better answer.

Reading project connections in code

- **project.connections.list()** returns connections, but **never their credentials**.
- **connections.get(name)** returns one connection, including its target URL.
- Credentials come back only if you pass **include_credentials=True**. The default is off.
- **connections.get_default(type)** finds the default connection of a type, such as AI Search.
- Older samples build a client from a **project connection string**. That is the hub-era pattern.

> ON THE EXAM
>
> Prefer Entra ID on the connected service over fetching its keys. include_credentials is for resources that only accept keys.

Connection errors and their causes

SYMPTOM	MOST LIKELY CAUSE
401 or 403 with Entra ID	Missing role, or token requested for the wrong scope
404 Not Found	model isn't a deployment name, or wrong endpoint URL
400 with code content_filter	A guardrail blocked the prompt
429 Too Many Requests	Rate limit or capacity: see module 06
HTTP 200, but the answer stops mid-stream	An error event inside the stream

THE SIGNAL

404 is the classic first-day error: the code names the model, not the deployment.

> A developer deploys gpt-5-mini in a Foundry project under the deployment name chat-prod. Their app authenticates with DefaultAzureCredential and holds the Foundry User role.
>
> Every call to responses.create with model set to gpt-5-mini returns 404 Not Found.

What should the developer change?

A Assign the Foundry Account Owner role on the Foundry resource.

B Set model to chat-prod, the deployment's name, in the request.

C Add a dated api-version query parameter to every request URL.

D Build the client from the project's connection string instead.

◆ PAUSE AND DECIDE

What should the developer change?

A Foundry Account Owner.

Permission problems return 401 or 403, not 404. This role can't even build in a project.

B Use the deployment name.

The model parameter must match a deployment name. gpt-5-mini names no deployment here.

C Add an api-version.

The v1 API doesn't need dated versions. Adding one won't create a missing deployment.

D Use a connection string.

An older hub-era pattern. The endpoint isn't the fault.

THE SIGNAL

404 points to a name that doesn't exist. In Foundry, that name is almost always the deployment.

A healthcare app uses a reasoning model over several conversation turns. Compliance states that the AI service itself must not retain any conversation data.

The team still needs the model's reasoning context to carry across turns for accurate answers.

How should the team call the Responses API?

A Chain turns with previous_response_id, then delete responses after each session.

B Run every turn in background mode and poll until each response completes.

C Set store to false and pass encrypted reasoning items back with each turn.

D Use Chat Completions with store set to true, and export the stored history.

◆ PAUSE AND DECIDE

How should the team call the Responses API?

A previous_response_id, then delete.

Data is still stored until deleted, which breaks 'must not retain'.

B Background mode.

Background mode requires store=true, the opposite of the requirement.

C store=false with encrypted reasoning.

Nothing is retained by the service, and reasoning context still carries across turns.

D Chat Completions with store=true.

Explicitly stores the conversations the policy forbids.

THE SIGNAL

'Must not retain' rules out anything that stores first and deletes later. Stateless from the start is the only fit.

> An insurer sends 300-page scanned claim files as PDF input to a vision model, to pull out three fields: claim number, date and amount.
>
> Results are accurate, but token costs are far above budget, and some requests approach input limits.

What should the team do?

A Stop sending whole files; extract fields with a document extractor.

B Upload each PDF with the Files API instead of sending it as base64.

C Switch to a text-only model, which is cheaper per token for PDFs.

D Raise max_output_tokens so the model can answer in a single request.

◆ PAUSE AND DECIDE

What should the team do?

A Use a document extractor instead.

Every page adds text plus an image to context. Fixed fields suit a purpose-built extractor.

B Upload via the Files API.

Changes how the file is sent, not what enters context. Every page still costs tokens.

C Text-only model.

Only models accepting text and images can take PDF input at all.

D Raise max_output_tokens.

The cost is in the input. Output length isn't the problem.

THE SIGNAL

'Three fixed fields from long documents' is a modules 02 and 32 signal: a purpose-built extractor beats a general model.

> A retail chatbot streams answers with the Responses API. The operations dashboard shows every request succeeding with HTTP 200.
>
> Yet users report that some answers stop part-way through a sentence during busy periods.

What should the developers do?

A Treat it as normal model behaviour and add a button to regenerate.

B Switch the chatbot from the Responses API to Chat Completions.

C Retry any request whose HTTP status code is anything other than 200.

D Handle error events in the stream, then retry or tell the user.

◆ PAUSE AND DECIDE

What should the developers do?

A Normal behaviour; add regenerate.

It isn't normal. A mid-stream error occurred and the app ignored it.

B Switch to Chat Completions.

Doesn't address the cause, and loses the recommended API's features.

C Retry non-200 statuses.

Every response is 200. This check never fires.

D Handle stream error events.

Mid-stream failures arrive as error events inside a 200. Detect them and react.

THE SIGNAL

'All 200s, but answers stop' plus 'busy periods' points to too_many_requests events inside the stream.

Connecting apps in five lines

- **Endpoint, credential, deployment name:** the three failure points of every call.
- **Responses API** for new work. The v1 API needs no dated api-version.
- **store=false** when nothing may be retained. Background mode needs storage.
- **Streaming failures arrive inside HTTP 200.** PDFs add every page as text and image.
- **connections.list** never returns credentials. 404 usually means the deployment name.

ON THE EXAM

Next, module 12: tuning generation with prompts, parameters, reasoning effort and structured output.